SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1596

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 5

Duration : 60 Minutes

Code of Conduct
I Aakash R(192524462) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: K Sumanth

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-I

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

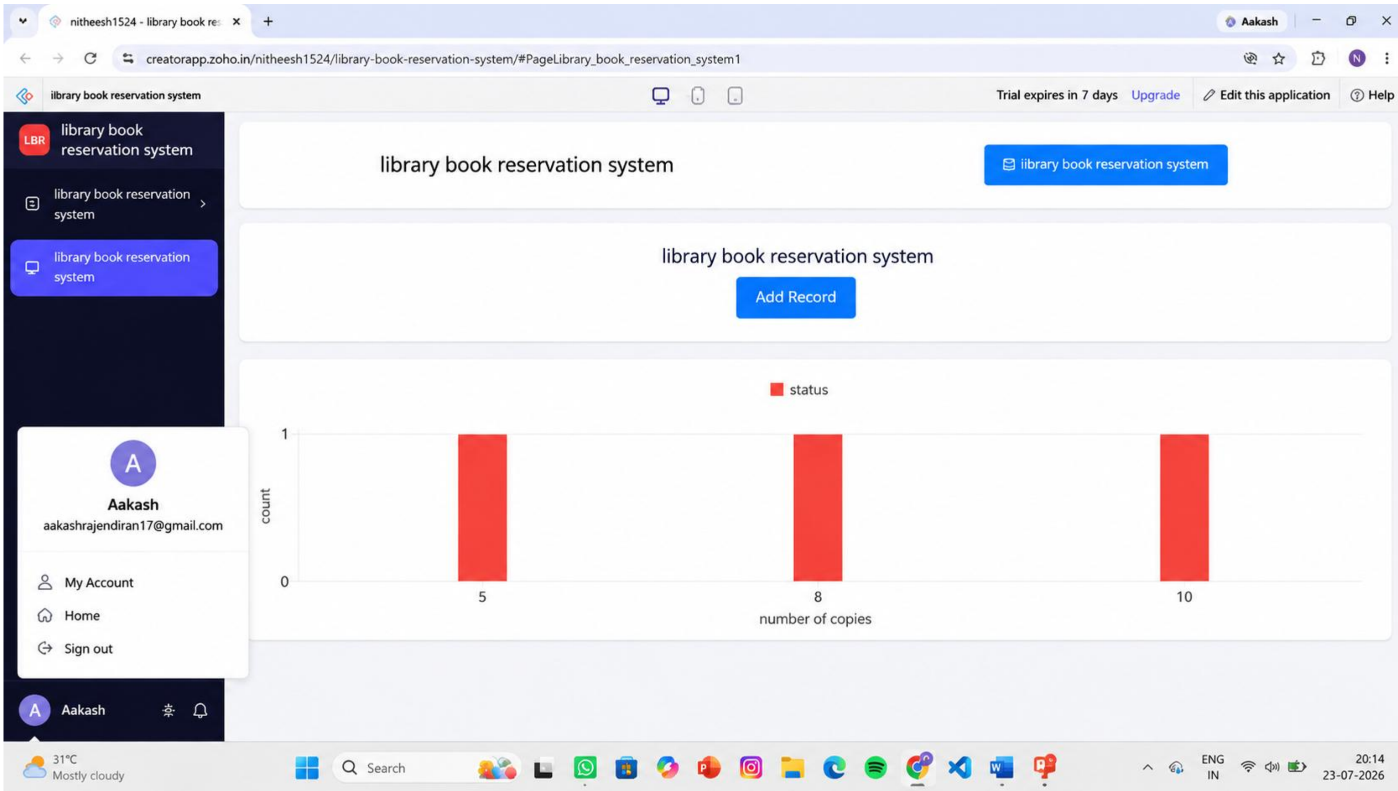
Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

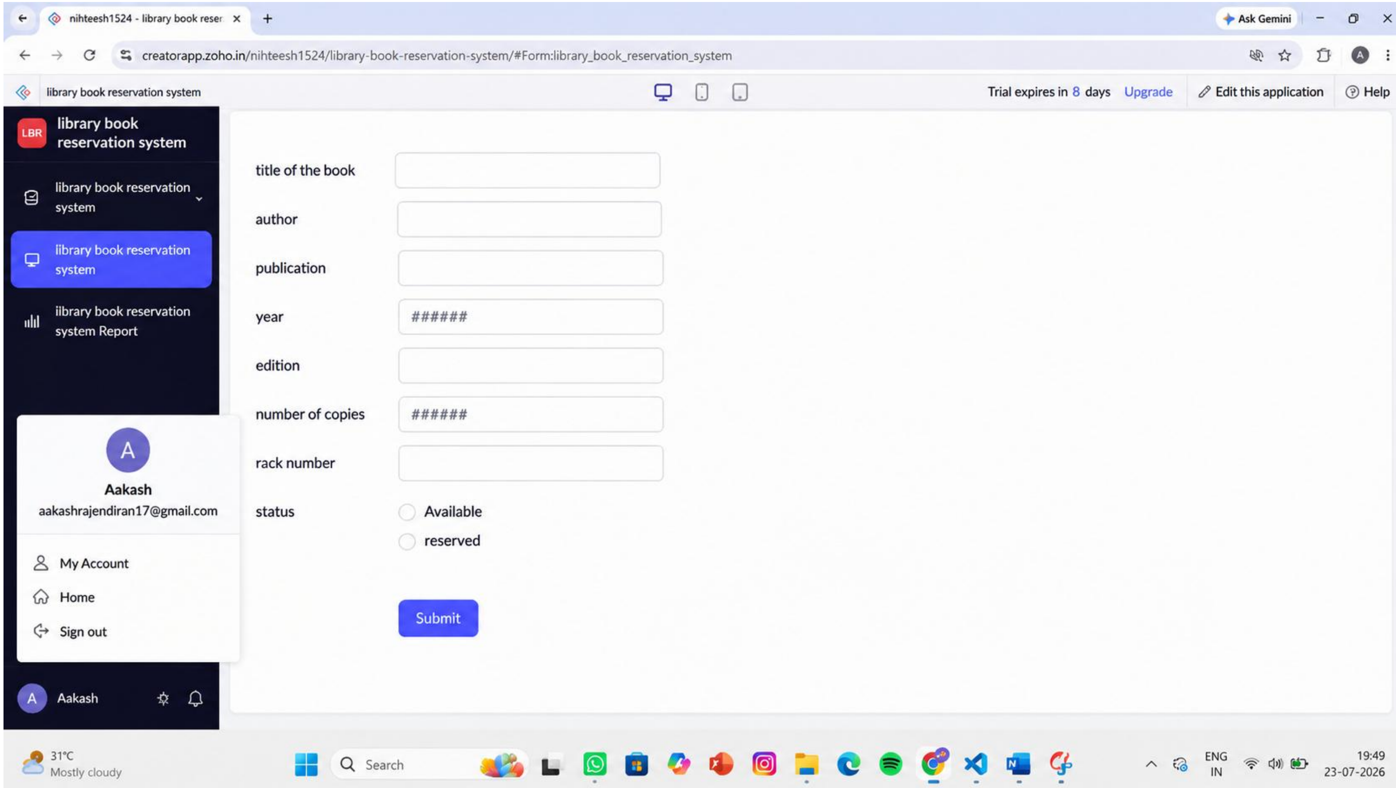
Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Library book reservation system

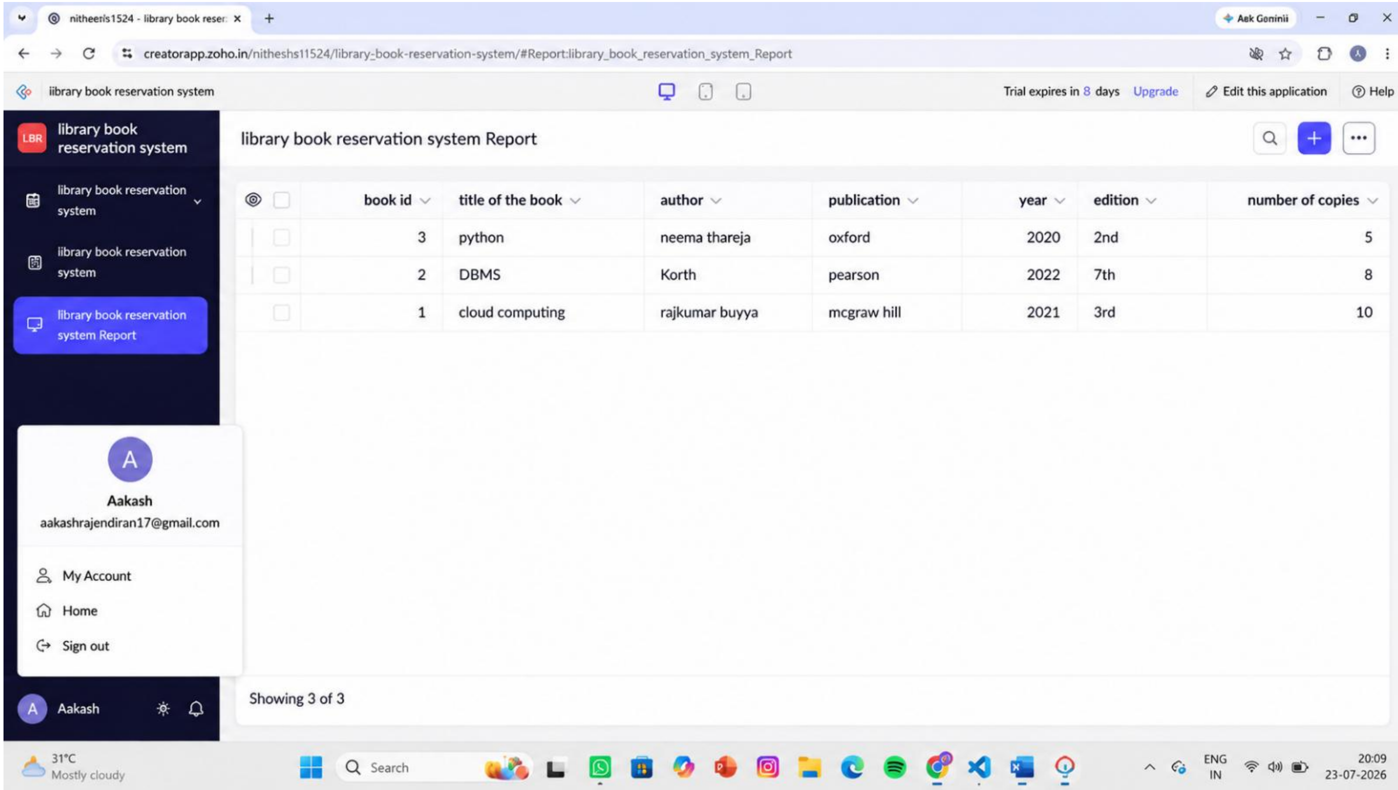
Dashboard



Form

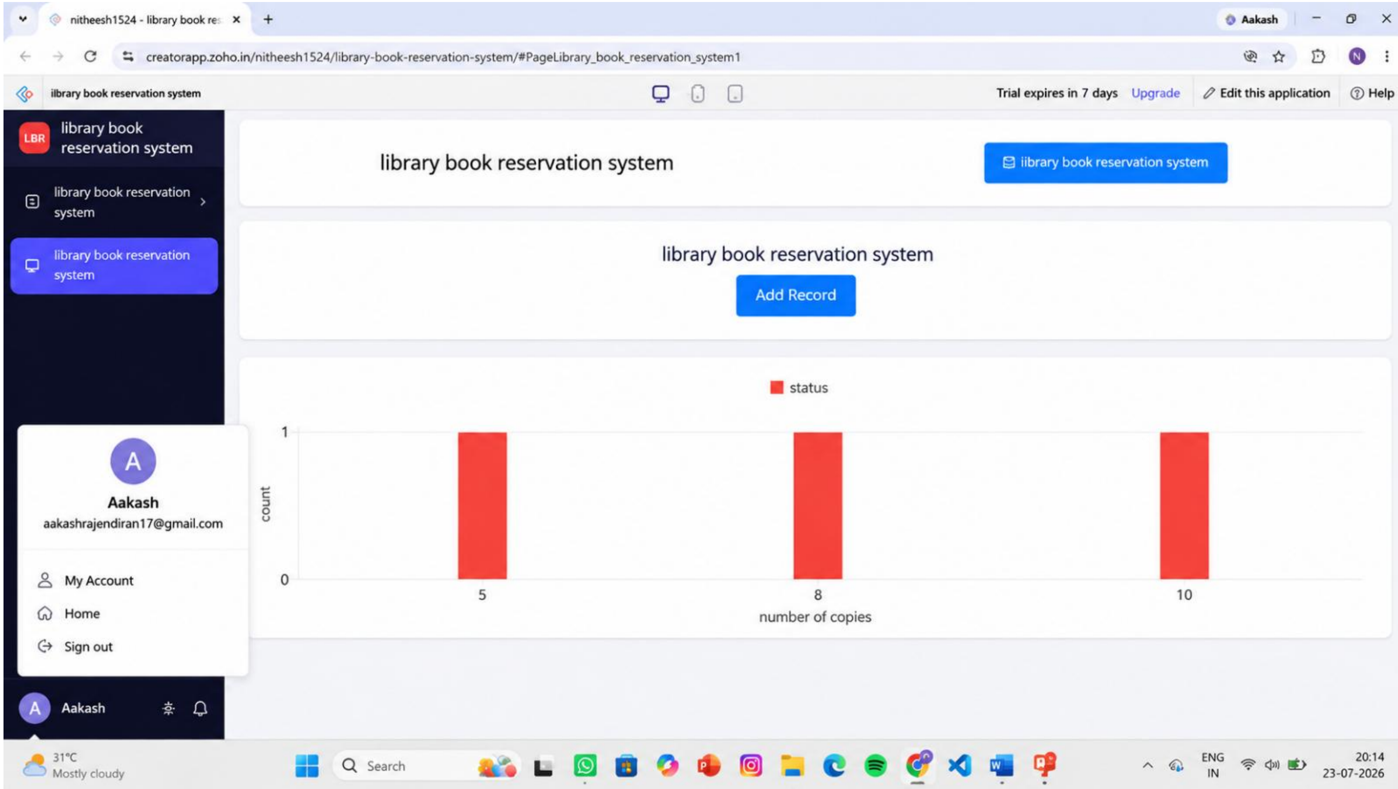

nihteesh1524 - library book reser
Ask Gemini
creatorapp.zoho.in/nihteesh1524/library-book-reservation-system/#Form:library_book_reservation_system
library book reservation system
Trial expires in 8 days
Upgrade
Edit this application
Help
library book reservation system
library book reservation system
library book reservation system
library book reservation system Report
Aakash
aakashrajendiran17@gmail.com
My Account
Home
Sign out
title of the book
author
publication
year
edition
number of copies
rack number
status
Available
reserved
Submit
31°C
Mostly cloudy
Search
ENG
IN
19:49
23-07-2026

Report

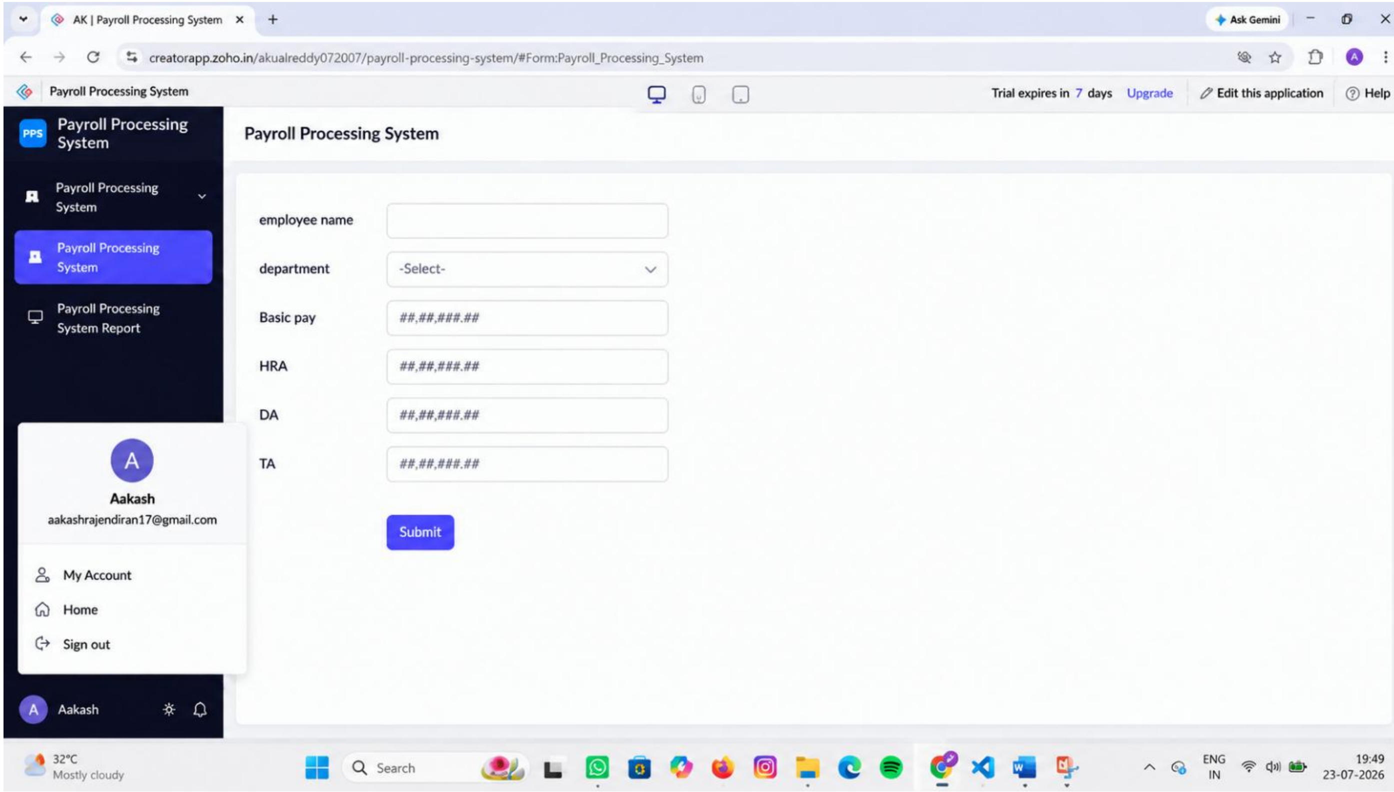


Experiment 2: Payroll Processing System

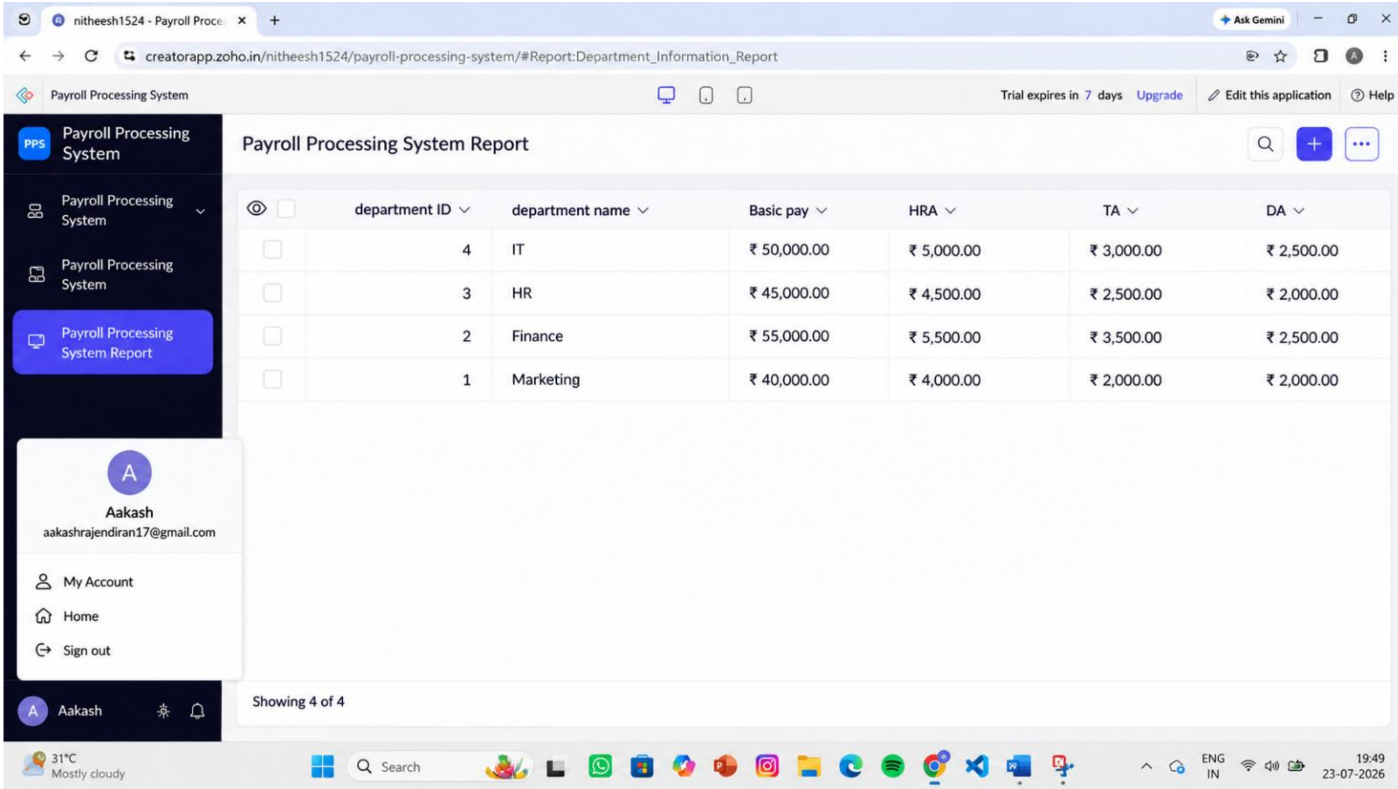
Dashboard



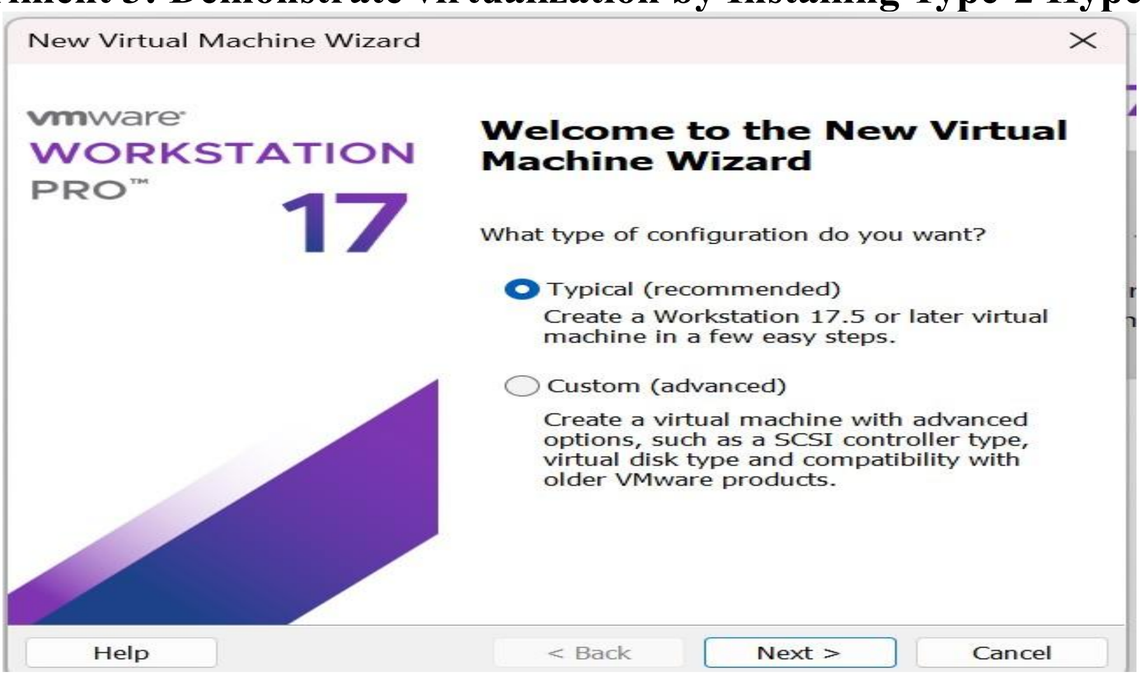
Form

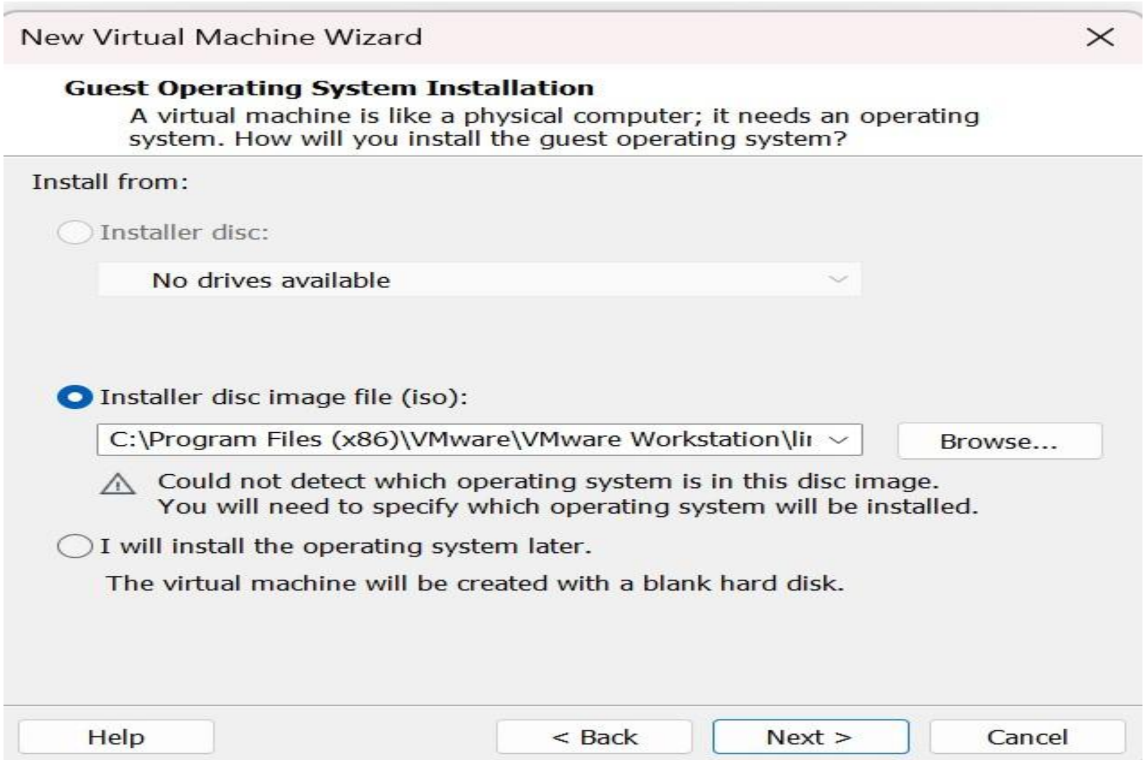

AK | Payroll Processing System
Ask Gemini
creatorapp.zoho.in/akualreddy072007/payroll-processing-system/#Form:Payroll_Processing_System
Payroll Processing System
Trial expires in 7 days
Upgrade
Edit this application
Help
PPS
Payroll Processing System
Payroll Processing System
Payroll Processing System
Payroll Processing System Report
Payroll Processing System
employee name
department
-Select-
Basic pay
##,##,###.##
HRA
##,##,###.##
DA
##,##,###.##
TA
##,##,###.##
Submit
A
Aakash
aakashrajendiran17@gmail.com
My Account
Home
Sign out
Aakash
32°C
Mostly cloudy
Search
ENG
IN
19:49
23-07-2026

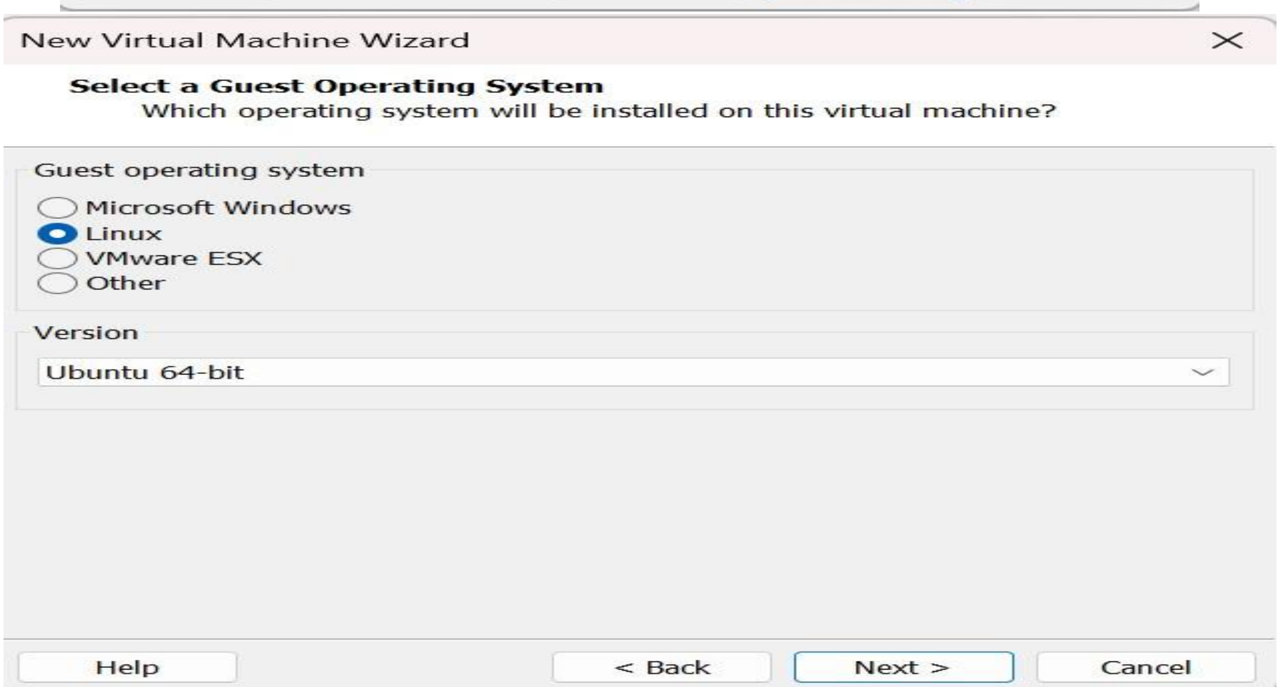
Report

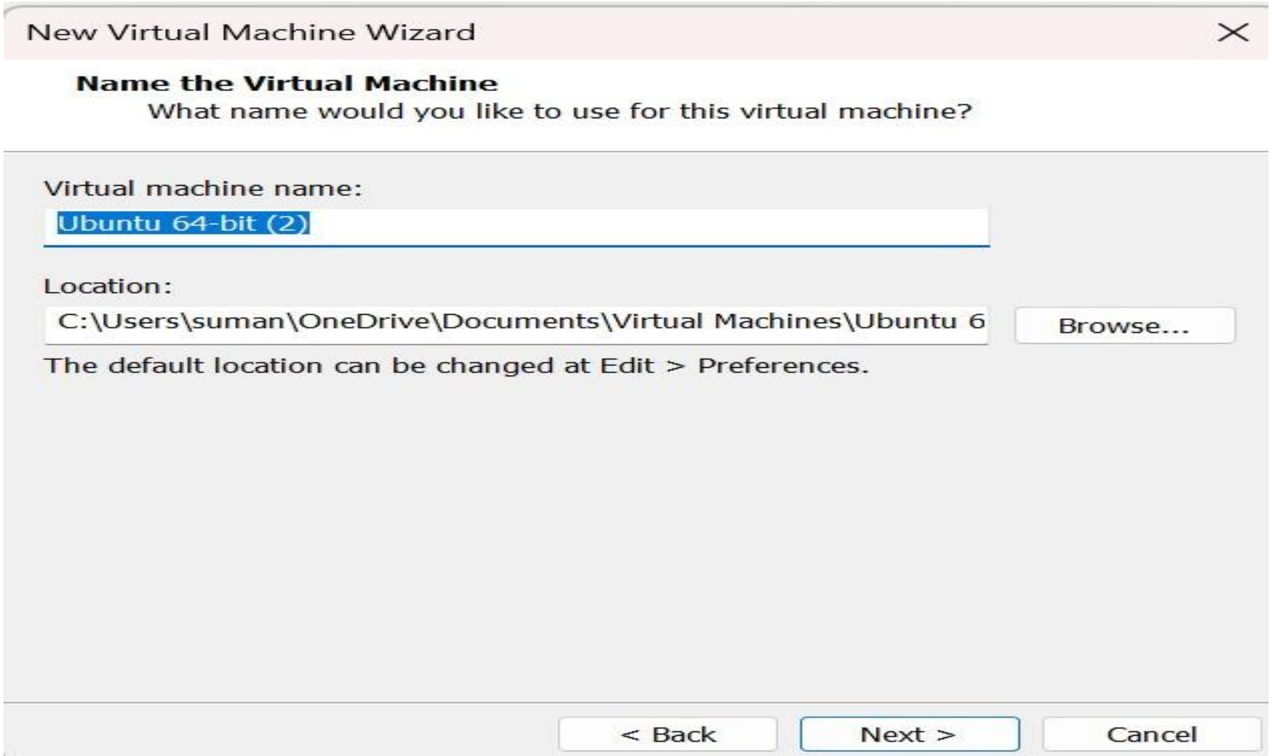


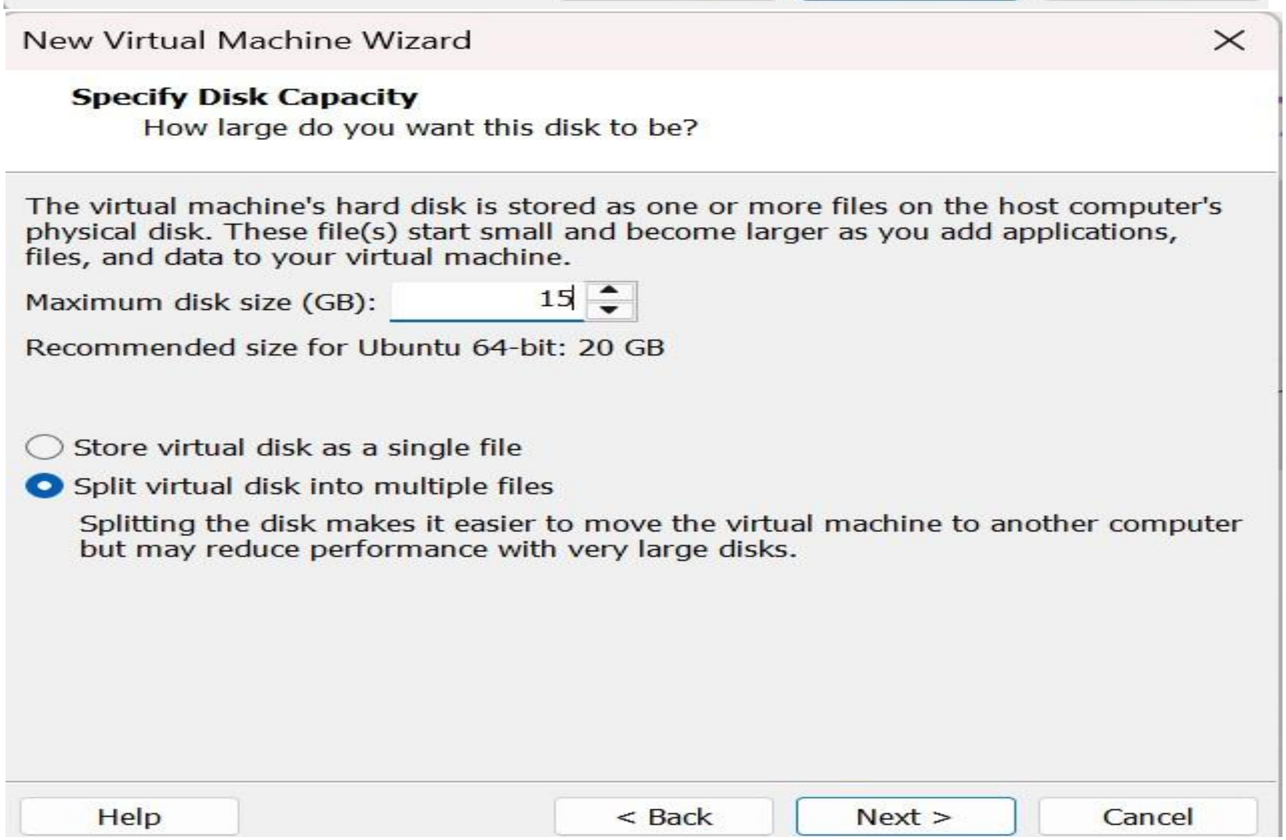
Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor

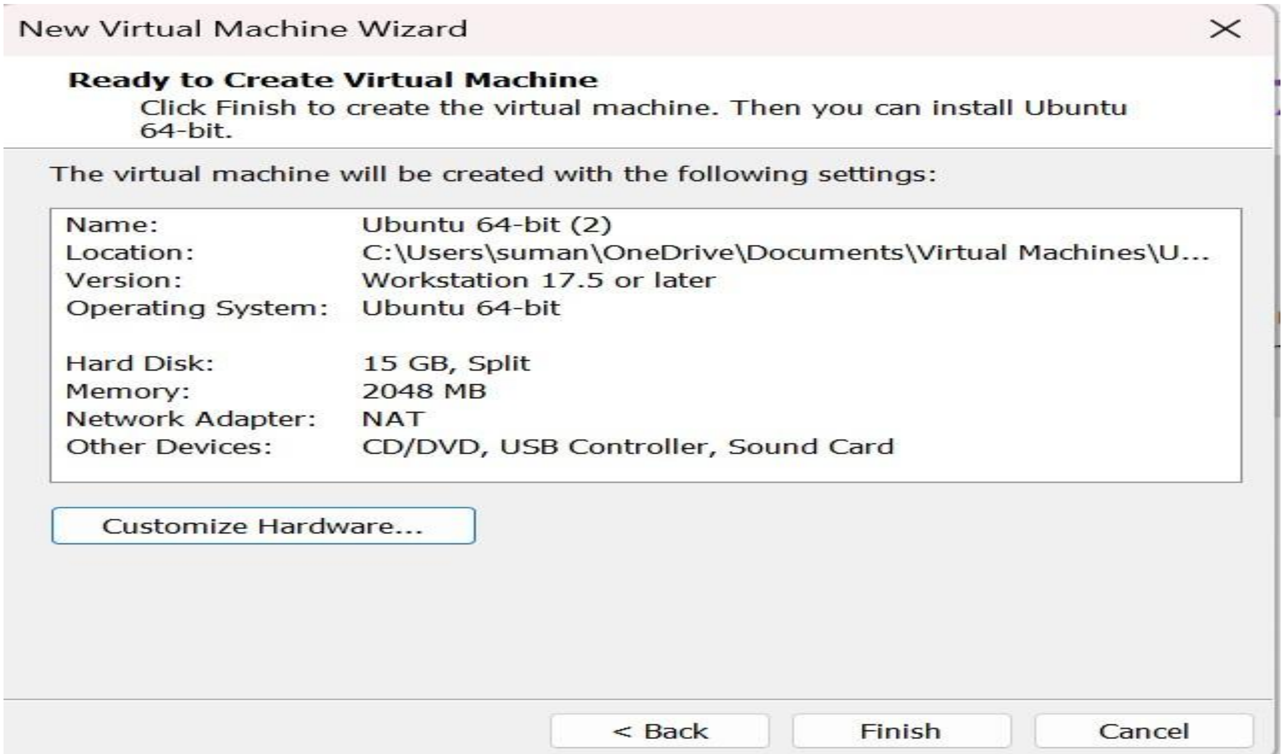



New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Program Files (x86)\VMware\VMware Workstation\lir
Browse...
Could not detect which operating system is in this disc image.
You will need to specify which operating system will be installed.
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

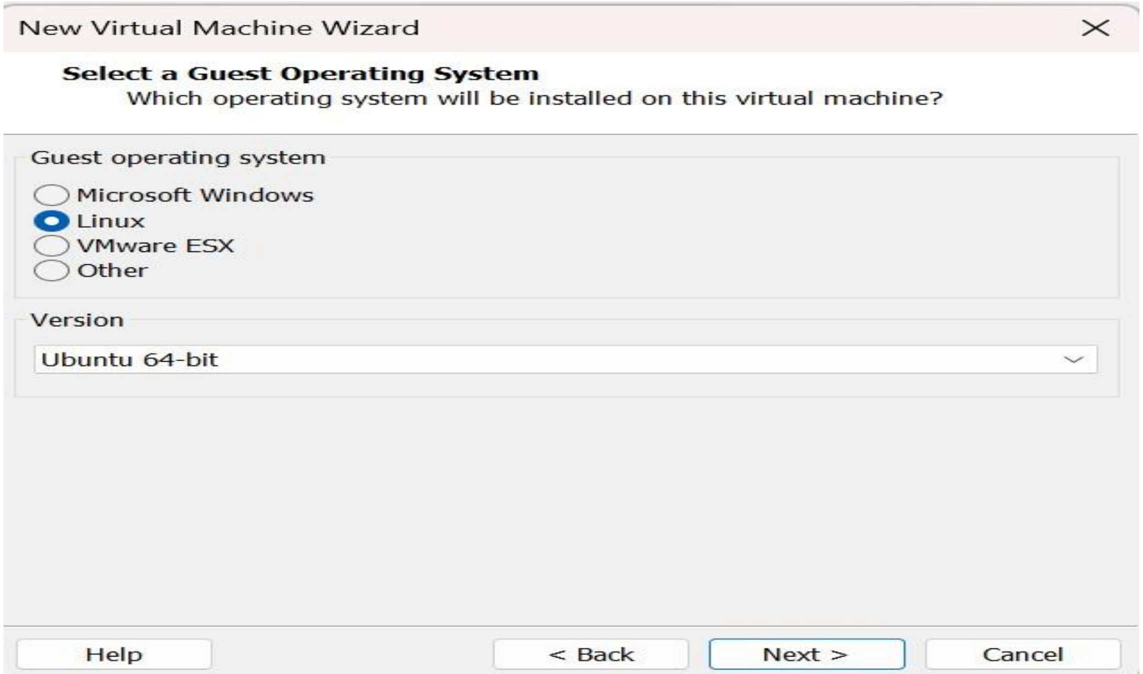

New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Ubuntu 64-bit
Help
< Back
Next >
Cancel

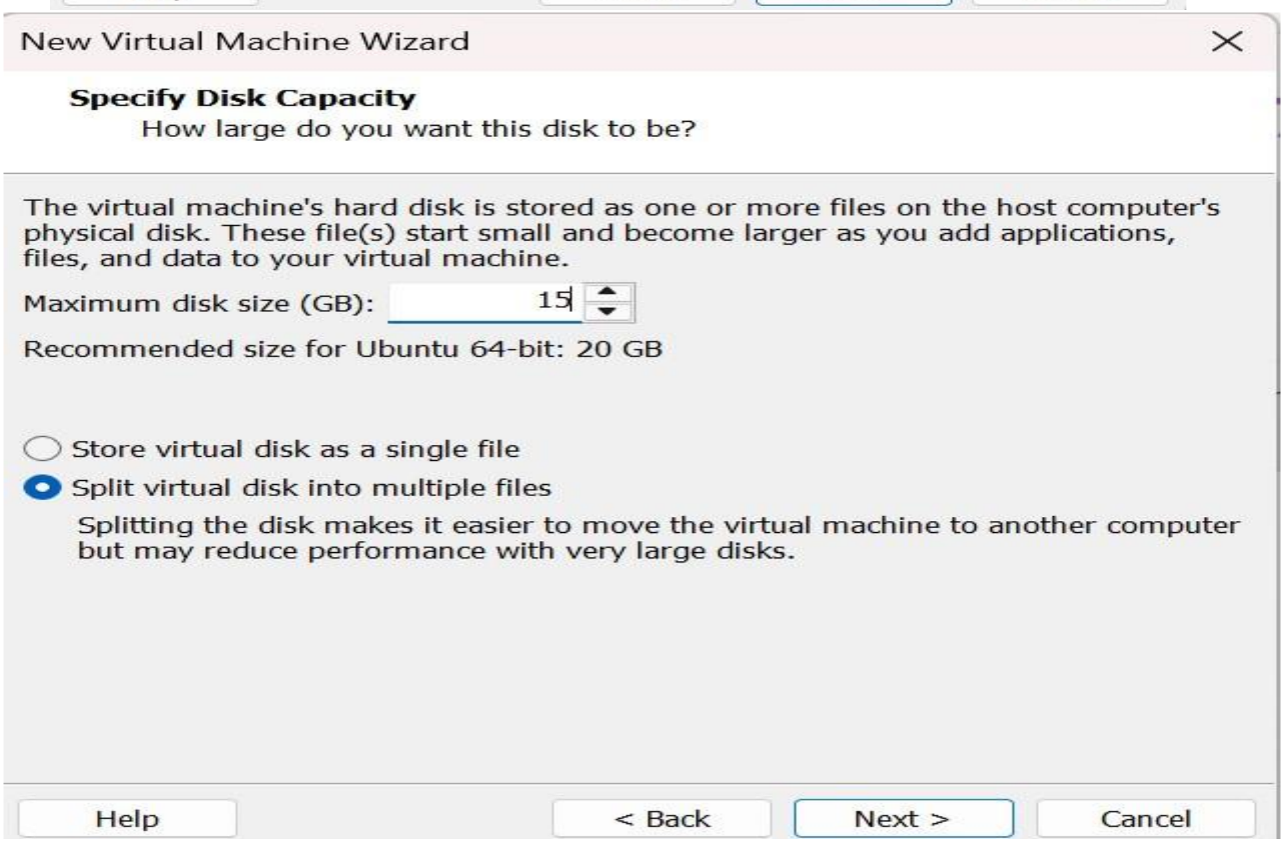

New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit (2)
Location:
C:\Users\suman\OneDrive\Documents\Virtual Machines\Ubuntu 6
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 15
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor


New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Ubuntu 64-bit
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
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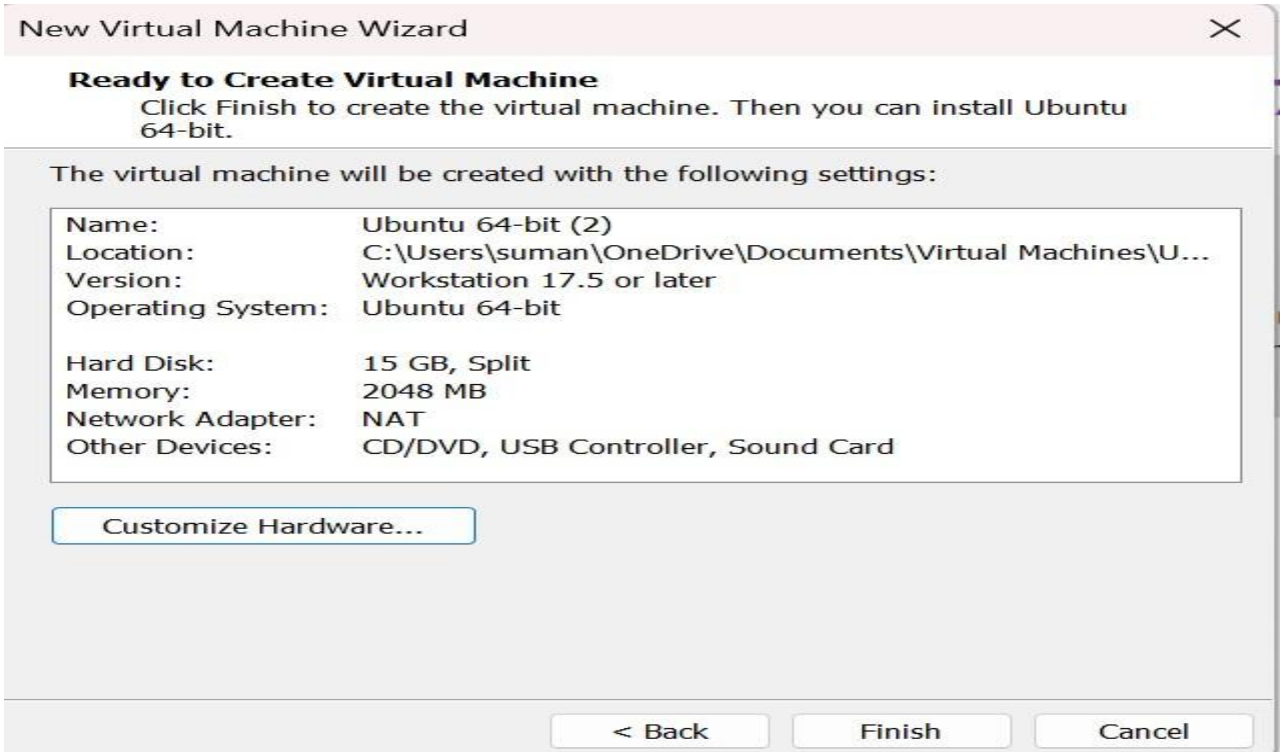
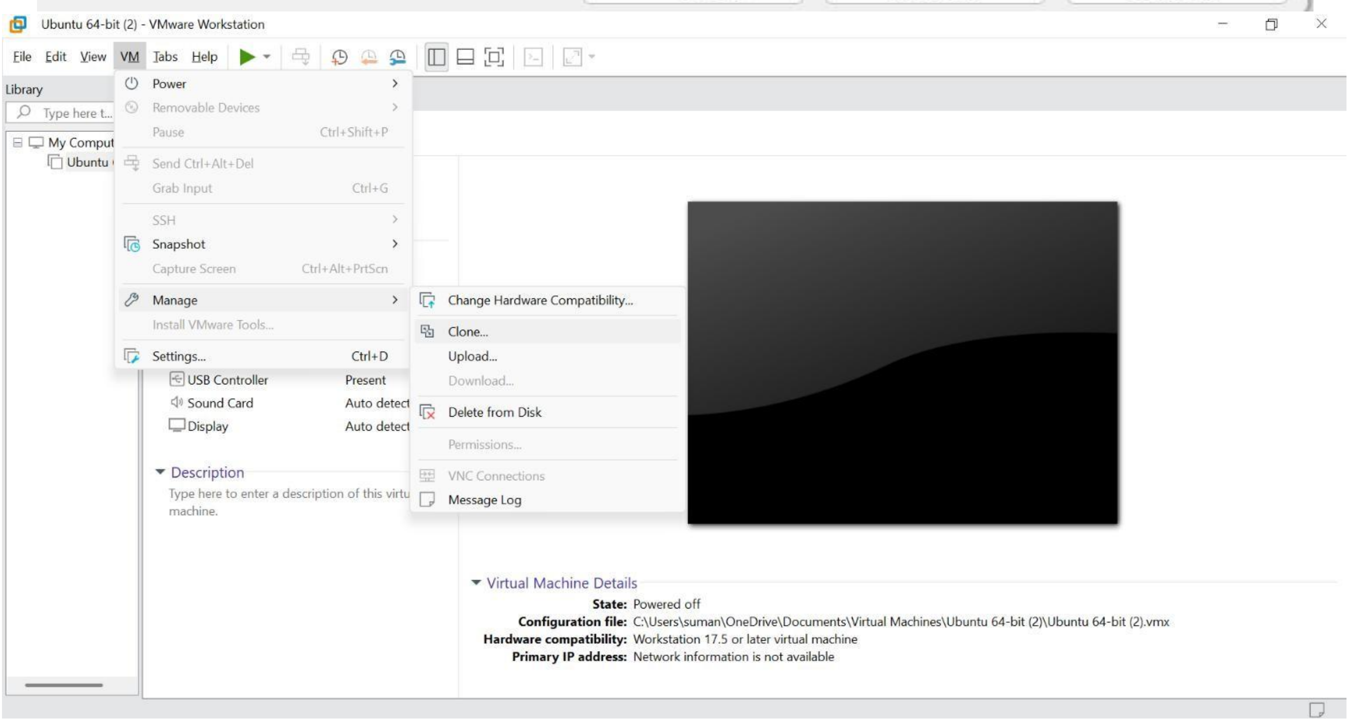
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.
The virtual machine will be created with the following settings:
Name: Ubuntu 64-bit (2)
Location: C:\Users\suman\OneDrive\Documents\Virtual Machines\U...
Version: Workstation 17.5 or later
Operating System: Ubuntu 64-bit
Hard Disk: 15 GB, Split
Memory: 2048 MB
Network Adapter: NAT
Other Devices: CD/DVD, USB Controller, Sound Card
Customize Hardware...
< Back
Finish
Cancel


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here t...
My Comput
Ubuntu
Power
Removable Devices
Pause Ctrl+Shift+P
Send Ctrl+Alt+Del
Grab Input Ctrl+G
SSH
Snapshot
Capture Screen Ctrl+Alt+PrtScn
Manage
Install VMware Tools...
Settings... Ctrl+D
Change Hardware Compatibility...
Clone...
Upload...
Download...
Delete from Disk
Permissions...
VNC Connections
Message Log
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtu machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\suman\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available